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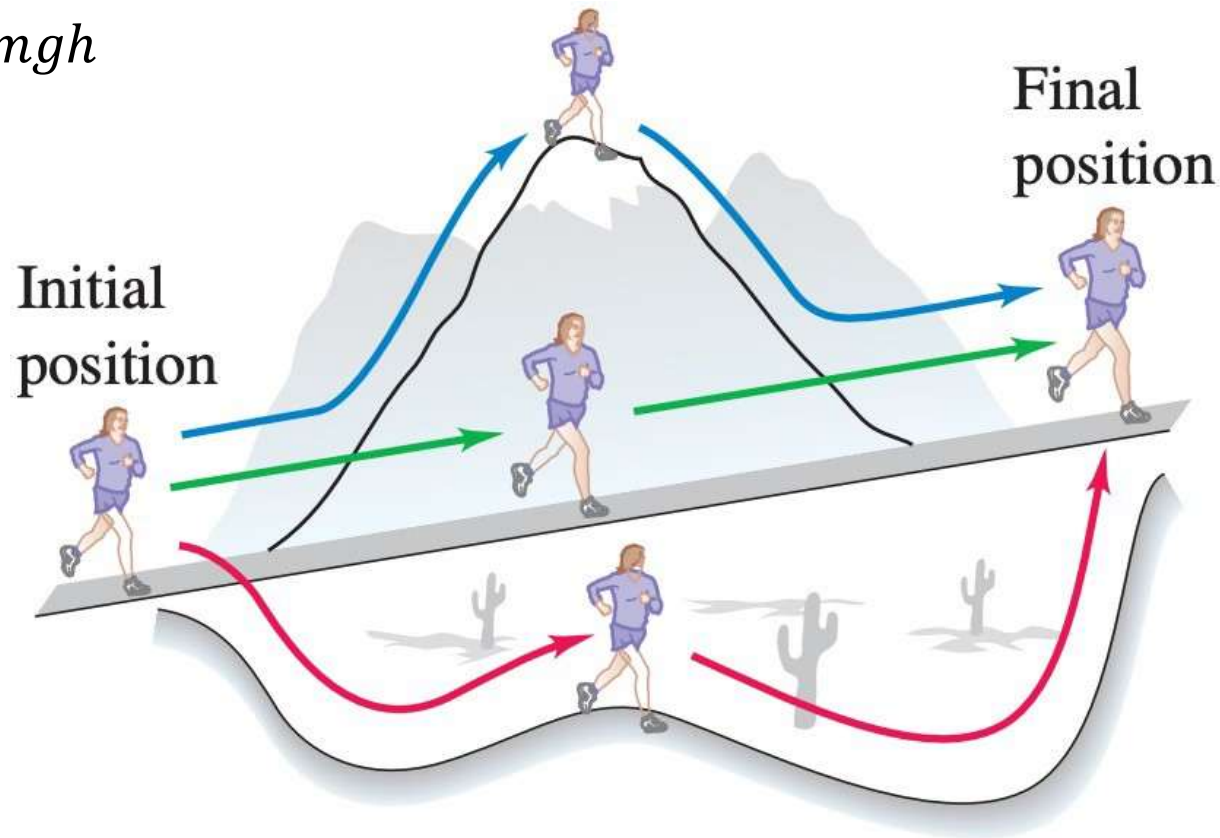
DTU Energy
B301, R214

Recap on Energy Conservation

Conservative vs nonconservative forces

Gravitational potential energy

$$U = mgh$$



Frictional losses

$$W_f = F_f \Delta s$$

Conservative vs. nonconservative forces

A conservative force is one in which the work is reversible (i.e. there are no losses), and is a difference between initial and final states of a potential energy function (i.e. it's independent of the path taken)

A nonconservative force is essentially everything that's not a conservative force

Example: Gravity is a conservative force, but friction is nonconservative

Force and potential energy:

Because force and potential energy are related: $\Delta U = F\Delta x$

We can express force in terms of the potential energy: $F = \frac{\Delta U}{\Delta x}$

If we then start considering small changes, we can write:

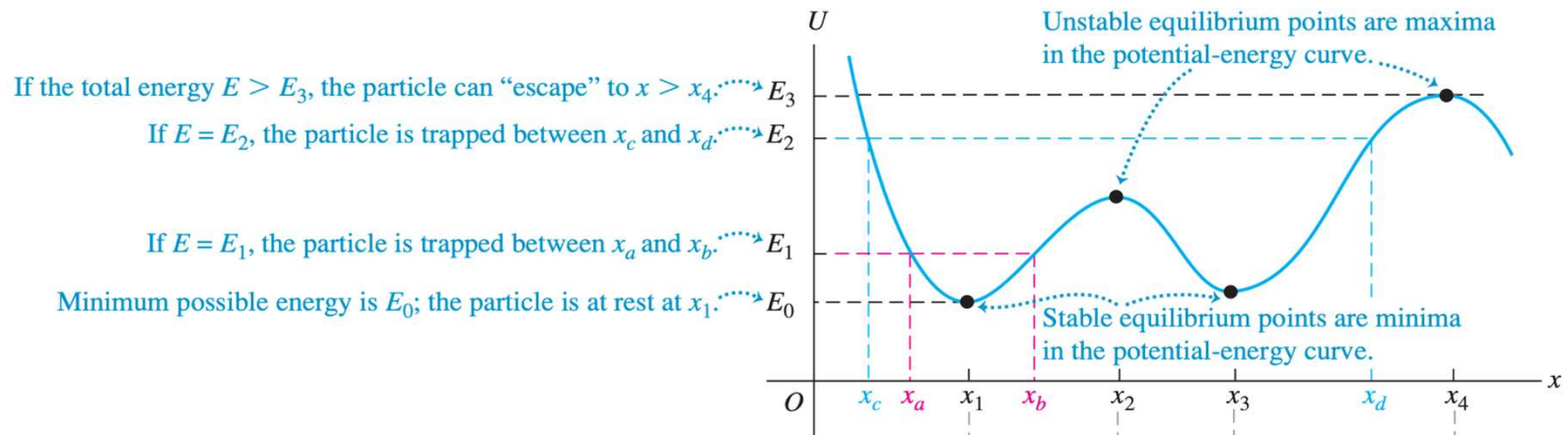
$$F = \frac{dU}{dx}$$

$$U = \int_{x_1}^{x_2} F dx$$

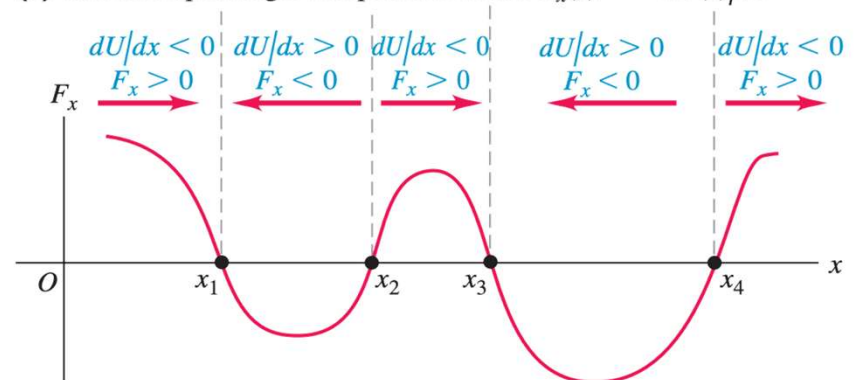
This means we can describe the energy functions associated with forces..

Energy diagrams

(a) A hypothetical potential-energy function $U(x)$



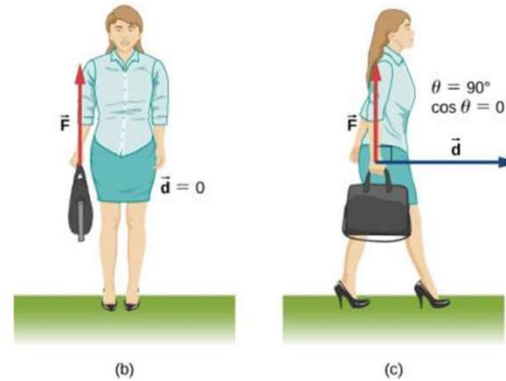
(b) The corresponding x -component of force $F_x(x) = -dU(x)/dx$



Question 1 (1 point)

You are moving a box in five steps:

1. you're lifting the box up
2. you accelerate it to a natural speed
3. you carry the box while walking with constant speed
4. you brake until you come to rest
5. you're slowing lowering the box until it rests at the floor.

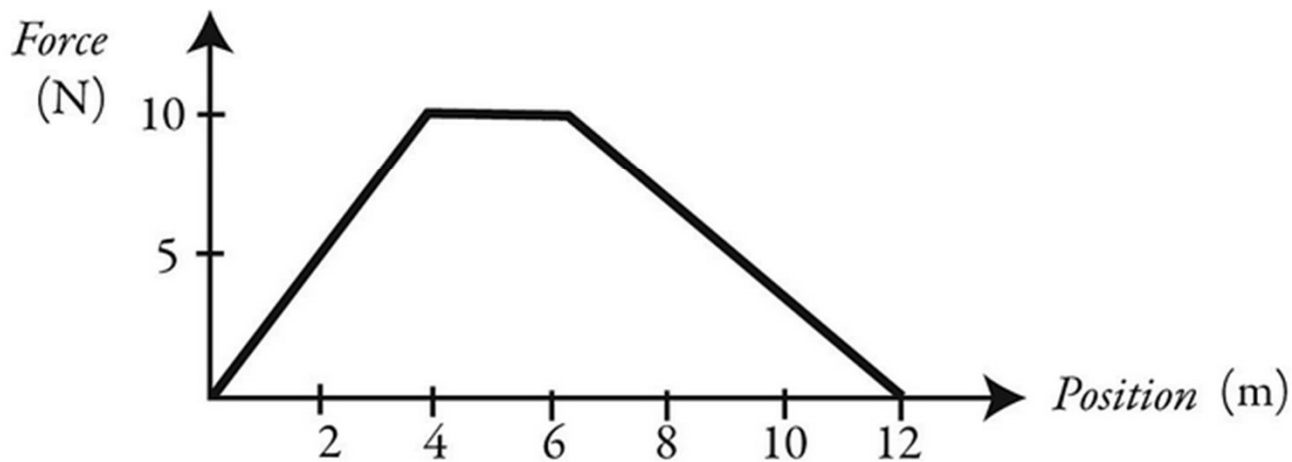


- ☐ Only in steps 1 and 5.
- ☐ Only in step 1.
- ☐ Only in steps 1, 2, 4, and 5.
- ☐ Only in steps 1 and 2.
- ☐ Only in steps 2 and 3.

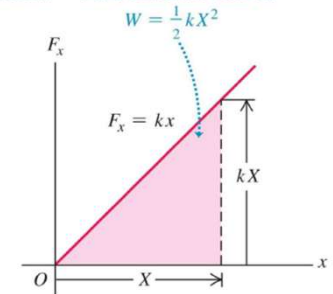
In which of the five steps do you perform positive work on the box?

Question 2 (1 point)

An object is subject to a force that varies with position as shown in the graph below.



The area under the graph represents the work done on the spring as the spring is stretched from $x = 0$ to a maximum value X :



Determine the work that the force performs in the interval
 $6 \text{ m} \leq x \leq 12 \text{ m}$

Determine the work that the force performs in the interval
 $0 \text{ m} \leq x \leq 6 \text{ m}$

Determine the work that the force performs in the interval
 $0 \text{ m} \leq x \leq 4 \text{ m}$

Question 3 (1 point)

The work, W_1 , that is required to accelerate a car from 0 km/h to 50 km/h is

- ☐ the same as the work, W_2 , that is required to accelerate the car from 50 km/h to 100 km/h.
- ☐ larger than the work, W_2 , that is required to accelerate the car from 50 km/h to 100 km/h.
- ☐ smaller than the work, W_2 , that is required to accelerate the car from 50 km/h to 100 km/h.
- ☐ can be smaller or larger or the same as the work, W_2 , that is required to accelerate the car from 50 km/h to 100 km/h depending on how long time the acceleration takes.

Question 4 (1 point)

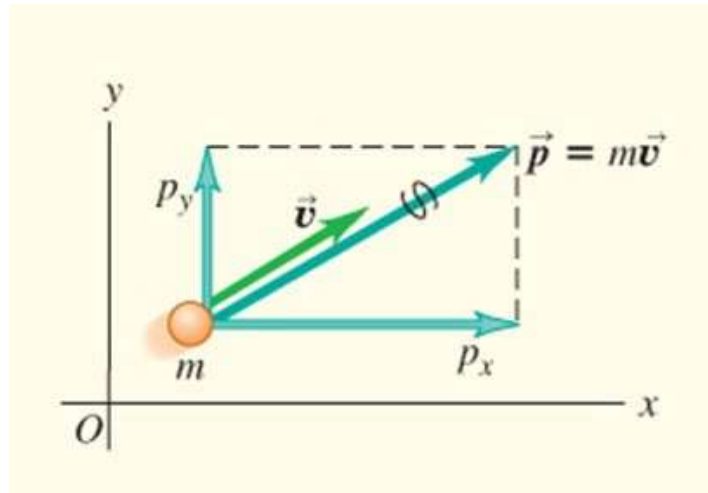
A ball with mass 3.00 kg is fixed to a string and being rotated in a vertical circular motion with radius 2.00 m. The string is all the time straight and is orthogonal to the velocity of the ball.

When the ball rotates from its lowest position to its highest position

- ☐ gravity performs the work 118 J on the ball and the tension performs the work 118 J on the ball.
- ☐ gravity performs the work -118 J on the ball and the tension performs the work -118 J on the ball.
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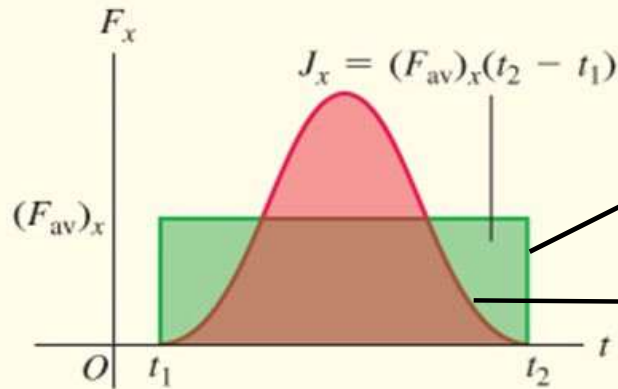
Collisions and Conservation of Linear Momentum

Linear momentum - definition



Newton's second law: If there is a net external force on an object ... $\sum \vec{F} = m\vec{a}$ Mass of object $\rightarrow \sum \vec{F} = m \frac{d\vec{v}}{dt} = \frac{d}{dt}(m\vec{v}) \rightarrow$ Newton's second law in terms of momentum: The net external force acting on a particle ... $\sum \vec{F} = \frac{d\vec{p}}{dt}$... equals the rate of change of the particle's momentum.

Impulse



Impulse of a constant net external force

$$\vec{J} = \sum \vec{F}(t_2 - t_1) = \sum \vec{F} \Delta t$$

Constant net external force
Time interval over which net external force acts

Impulse of a general net external force (either constant or varying)

$$\vec{J} = \int_{t_1}^{t_2} \sum \vec{F} dt$$

Upper limit = final time
Time integral of net external force
Lower limit = initial time

$$\vec{J} = \int_{t_1}^{t_2} \frac{d\vec{p}}{dt} dt$$

Newton's second law

in terms of momentum:

The net external force acting on a particle ...

$$\sum \vec{F} = \frac{d\vec{p}}{dt}$$

... equals the rate of change of the particle's momentum.

Impulse-momentum theorem: The impulse of the net external force on a particle during a time interval equals the change in momentum of that particle during that interval:

$$\vec{J} = \vec{p}_2 - \vec{p}_1 = \Delta \vec{p}$$

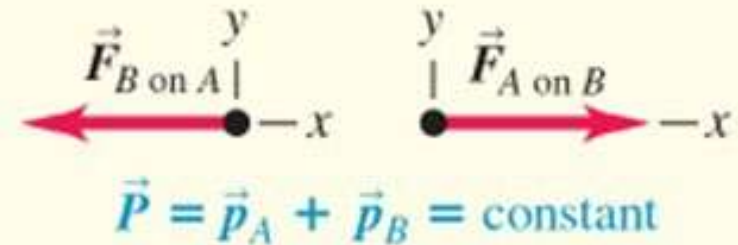
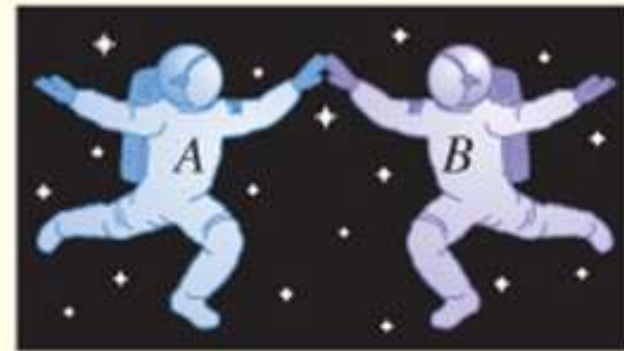
Impulse of net external force over a time interval
Final momentum
Initial momentum
Change in momentum

Conservation of momentum

$$\vec{P} = \vec{p}_A + \vec{p}_B + \dots$$

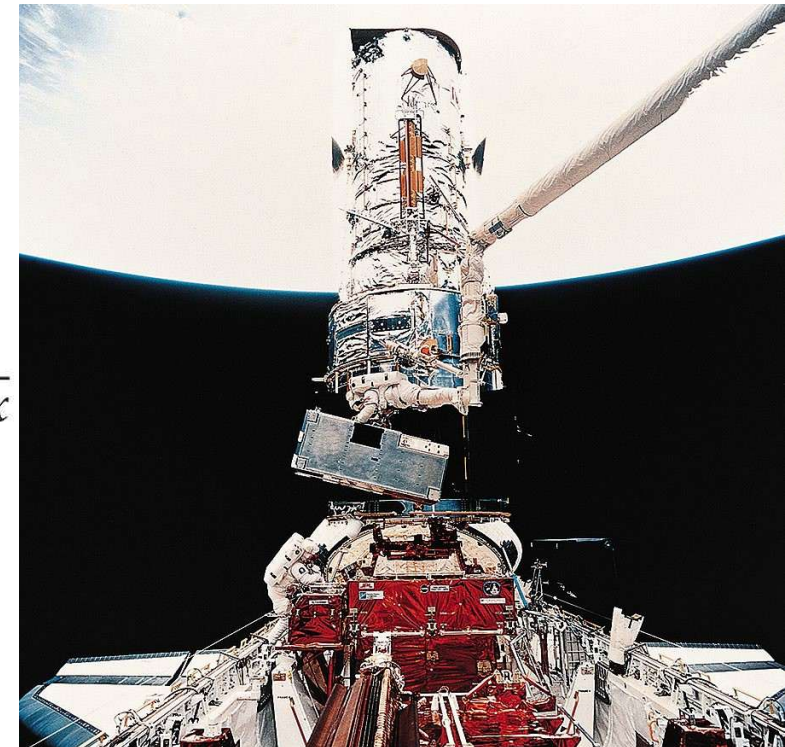
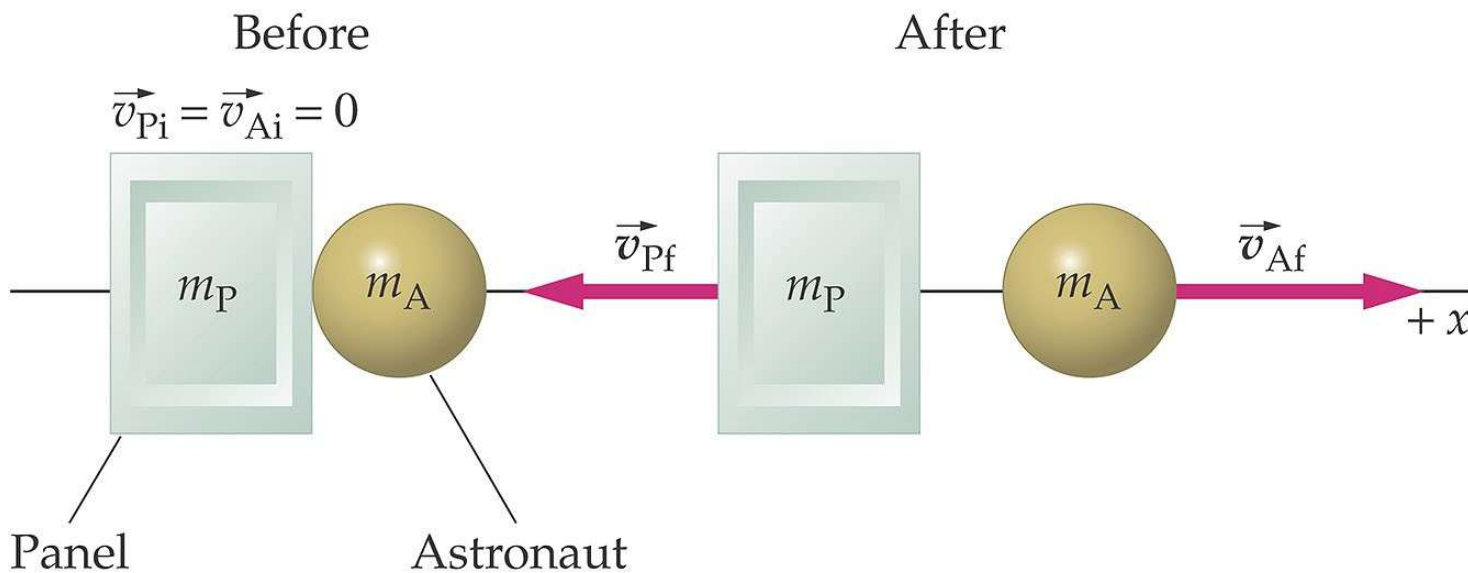
$$= m_A \vec{v}_A + m_B \vec{v}_B + \dots \quad (8.14)$$

If $\sum \vec{F} = \mathbf{0}$, then $\vec{P} = \text{constant}$.



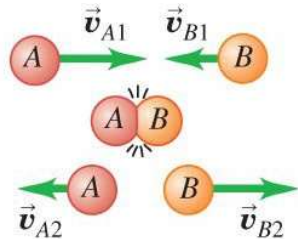
Example: astronaut replacing a solar panel at the Hubble telescope (pushing the detached panel away into space)

Discuss with your neighbor: draw a simple mechanical model for the situation
- before and after

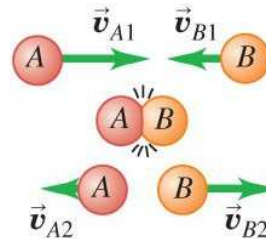


Types of collisions

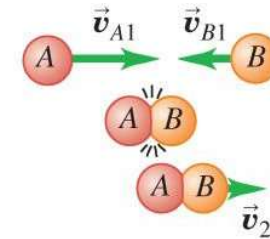
Elastic:
Kinetic energy conserved.



Inelastic:
Some kinetic energy lost.



Completely inelastic:
Objects have same final velocity.



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PROBLEM-SOLVING STRATEGY 8.1 Conservation of Momentum

IDENTIFY *the relevant concepts:* Confirm that the vector sum of the external forces acting on the system of particles is zero. If it isn't zero, you can't use conservation of momentum.

SET UP *the problem* using the following steps:

1. Treat each object as a particle. Draw "before" and "after" sketches, including velocity vectors. Assign algebraic symbols to each magnitude, angle, and component. Use letters to label each particle and subscripts 1 and 2 for "before" and "after" quantities. Include any given values.
2. Define a coordinate system and show it in your sketches; define the positive direction for each axis.
3. Identify the target variables.

EXECUTE *the solution:*

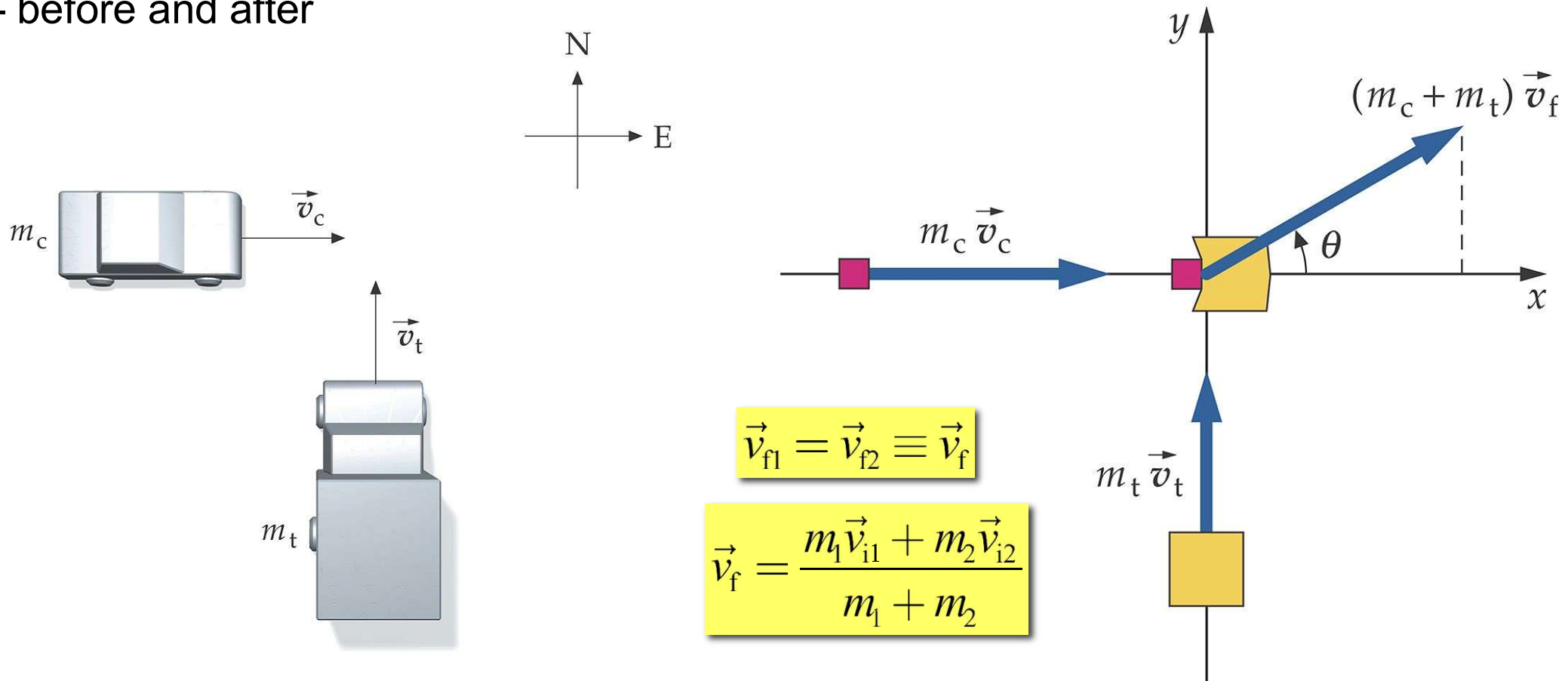
4. Write an equation in symbols equating the total initial and final x -components of momentum, using $p_x = mv_x$ for each particle. Write a corresponding equation for the y -components. Components can be positive or negative, so be careful with signs!
5. In some problems, energy considerations (discussed in Section 8.4) give additional equations relating the velocities.
6. Solve your equations to find the target variables.

EVALUATE *your answer:* Does your answer make physical sense? If your target variable is a certain object's momentum, check that the direction of the momentum is reasonable.

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Example: a car-truck collision – a totally inelastic collision

Discuss with your neighbor: draw a simple mechanical model for the situation
- before and after



Example: Exam August 2023

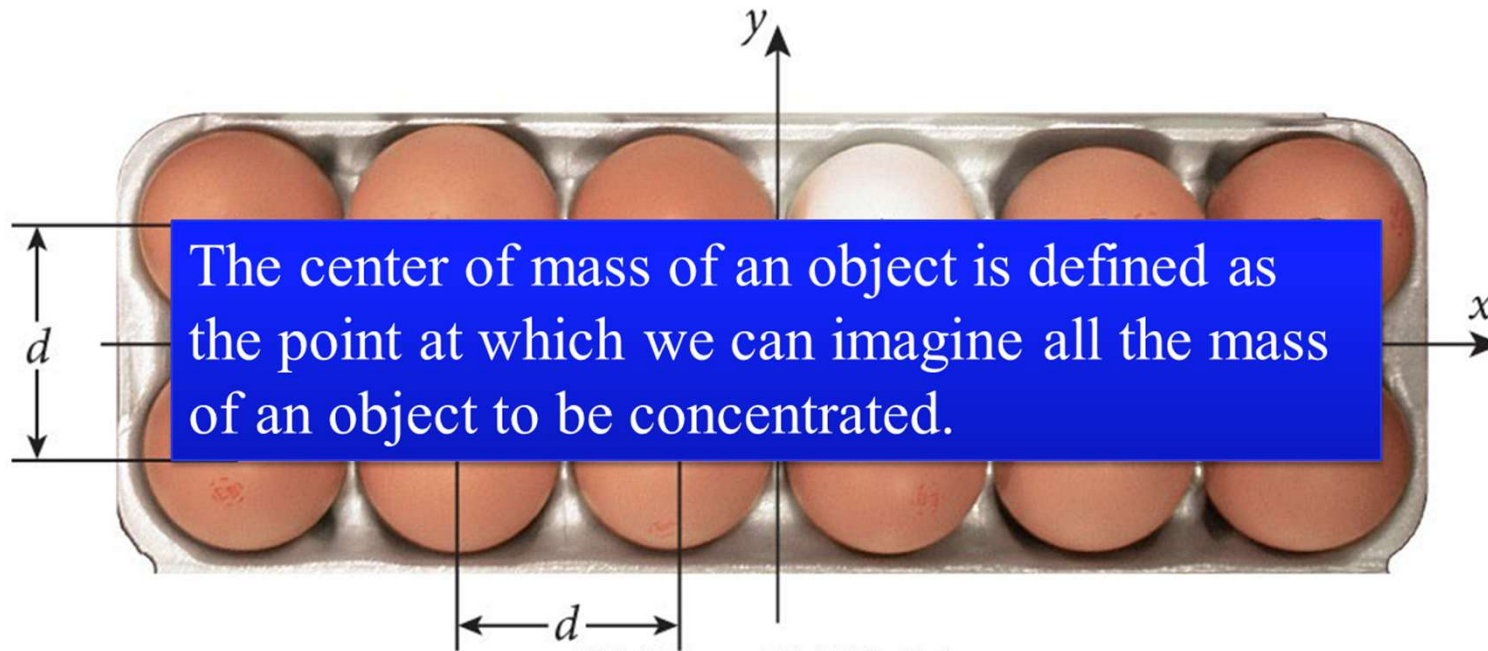
Two balls collide as shown in the figure below. Before the collision, the light ball is moving with horizontal velocity and the dark ball is at rest. After the impact, the dark sphere moves upwards to the right. The magnitude of the two velocities shown and the masses of the balls are unknown.



Which of the directions shown below for the speed of the light ball after impact are possible? The directions of the arrows are purely horizontal, purely vertical or oblique (the oblique arrows have no specific angle).

- A) ←
- B) →
- C) ↑
- D) ↓
- E) ↗
- F) ↖
- G) ↘
- H) ✓
- I) Do not know.

Center of Mass - Concept check



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Position vector of center of mass of a system of particles $\vec{r}_{cm}$

Position vectors of individual particles $\vec{r}_1, \vec{r}_2, \vec{r}_3, \dots$

Masses of individual particles $m_1, m_2, m_3, \dots$

$$\vec{r}_{cm} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + \dots}{m_1 + m_2 + m_3 + \dots} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$$

$$\vec{R} = \frac{1}{M} \int_V \vec{r} \rho(\vec{r}) dV$$

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Masses of individual particles $m_1, m_2, m_3, \dots$

$$\vec{r}_{\text{cm}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + \dots}{m_1 + m_2 + m_3 + \dots} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$$

$$\vec{v}_{\text{cm}} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2 + m_3 \vec{v}_3 + \dots}{m_1 + m_2 + m_3 + \dots}$$

Total mass of a system of particles M

Velocity of center of mass $\vec{v}_{\text{cm}}$

Momenta of individual particles $m_1 \vec{v}_1, m_2 \vec{v}_2, m_3 \vec{v}_3, \dots$

Total momentum of system $\vec{P}$

$$M \vec{v}_{\text{cm}} = m_1 \vec{v}_1 + m_2 \vec{v}_2 + m_3 \vec{v}_3 + \dots = \vec{P}$$

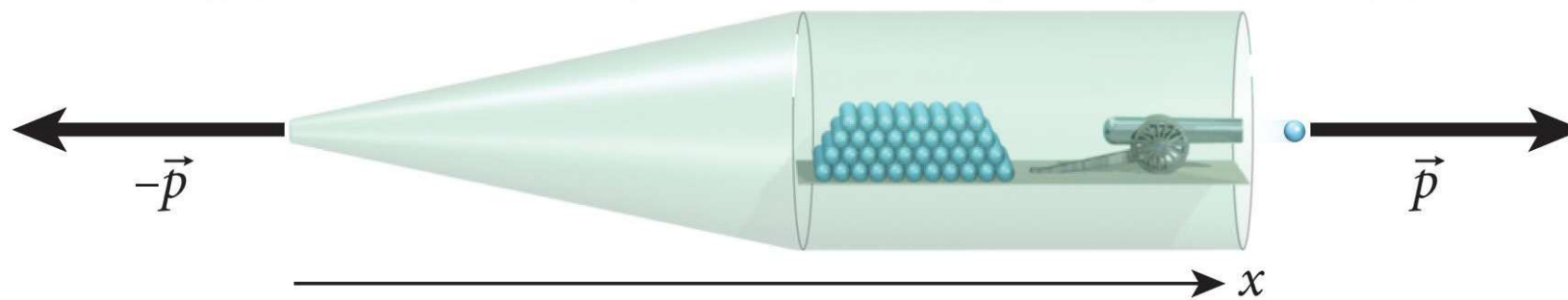
Total momentum of a system of particles A, B, C, ...

$$\vec{P} = \vec{p}_A + \vec{p}_B + \dots = m_A \vec{v}_A + m_B \vec{v}_B + \dots$$

... equals vector sum of momenta of all particles in the system.

$$\vec{P} = M \vec{v}_{\text{cm}} = \sum_{i=1}^n \vec{p}_i$$

Example: Rocket Motion



$$\vec{P} = M\vec{V}_{\text{CM}} = \sum_{i=1}^n \vec{p}_i$$

$$\frac{d}{dt} \vec{P} = \vec{F}_{\text{net}}$$

Net external force
on an object or a
collection of particles

$$\sum \vec{F}_{\text{ext}}$$

$$= M \vec{a}_{\text{cm}}$$

Total mass of object or
collection of particles

Acceleration of center of mass

The forces act upon CM. Only external
forces contribute to the net force.

Elastic Collisions in the CM reference frame

Internal forces do NOT change CM momentum

1. In the center-of-mass frame, total momentum is zero

Before the collision:

$$m_1 v_{1i}^* + m_2 v_{2i}^* = 0 \quad \vec{P} = M \vec{V}_{\text{CM}} = \sum_{i=1}^n \vec{p}_i$$

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2. In an elastic collision, kinetic energy is conserved

$$\frac{1}{2} m_1 (v_{1i}^*)^2 + \frac{1}{2} m_2 (v_{2i}^*)^2 = \frac{1}{2} m_1 (v_{1f}^*)^2 + \frac{1}{2} m_2 (v_{2f}^*)^2$$

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$$\frac{1}{2} \left(m_1 + \frac{m_1^2}{m_2} \right) (v_{1i}^*)^2 = \frac{1}{2} \left(m_1 + \frac{m_1^2}{m_2} \right) (v_{1f}^*)^2$$

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$$(v_{1i}^*)^2 = (v_{1f}^*)^2$$

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2. In an elastic collision, kinetic energy is conserved

$$\frac{1}{2} m_1 (v_{1i}^*)^2 + \frac{1}{2} m_2 (v_{2i}^*)^2 = \frac{1}{2} m_1 (v_{1f}^*)^2 + \frac{1}{2} m_2 (v_{2f}^*)^2$$

$$\frac{1}{2} m_1 (v_{1i}^*)^2 + \frac{1}{2} m_2 \left(-\frac{m_1}{m_2} v_{1i}^* \right)^2 = \frac{1}{2} m_1 (v_{1f}^*)^2 + \frac{1}{2} m_2 \left(-\frac{m_1}{m_2} v_{1f}^* \right)^2$$

$$\frac{1}{2} m_1 (v_{1i}^*)^2 + \frac{1}{2} m_2 \left(\frac{m_1}{m_2} \right)^2 (v_{1i}^*)^2 = \frac{1}{2} m_1 (v_{1f}^*)^2 + \frac{1}{2} m_2 \left(\frac{m_1}{m_2} \right)^2 (v_{1f}^*)^2$$

$$\frac{1}{2} m_1 (v_{1i}^*)^2 + \frac{1}{2} \frac{m_1^2}{m_2} (v_{1i}^*)^2 = \frac{1}{2} m_1 (v_{1f}^*)^2 + \frac{1}{2} \frac{m_1^2}{m_2} (v_{1f}^*)^2$$

$$\frac{1}{2} \left(m_1 + \frac{m_1^2}{m_2} \right) (v_{1i}^*)^2 = \frac{1}{2} \left(m_1 + \frac{m_1^2}{m_2} \right) (v_{1f}^*)^2$$

$$(v_{1i}^*)^2 = (v_{1f}^*)^2$$

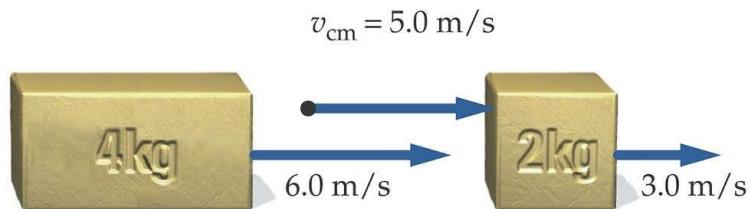
In the CM reference frame, particles simply change the sign of their velocities

$$v_{1f}^* = \pm v_{1i}^*$$

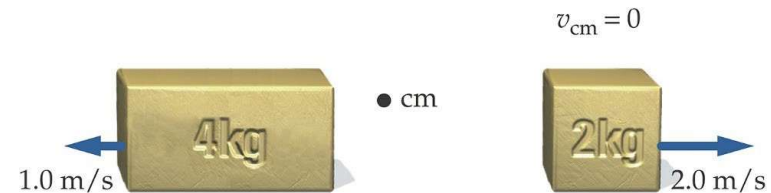
Example: Elastic Collisions in the CM reference frame

Internal forces do NOT change CM momentum

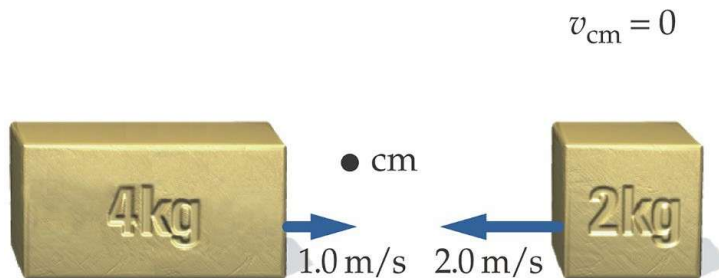
Initial conditions



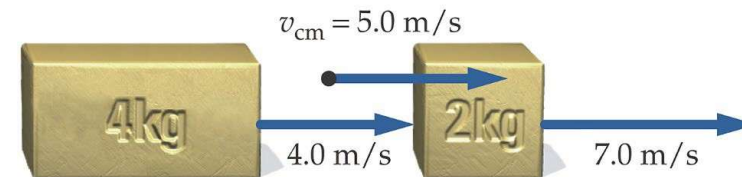
Solve collision



Transform to the center-of-mass frame by subtracting v_{cm}



Transform back to the original frame by adding v_{cm}



Summary: Momentum

Momentum

$$\vec{p} = m\vec{v}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

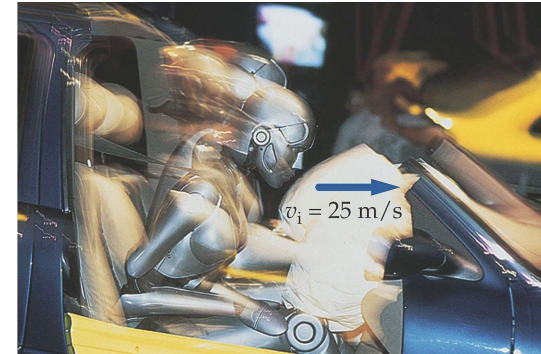
$$\vec{p}_{f1} + \vec{p}_{f2} = \vec{p}_{i1} + \vec{p}_{i2}$$

Impulse

$$\Delta\vec{p} \equiv \vec{p}_f - \vec{p}_i$$

$$\vec{J} = \vec{F}_{\text{ave}} \Delta t = m\Delta\vec{v}$$

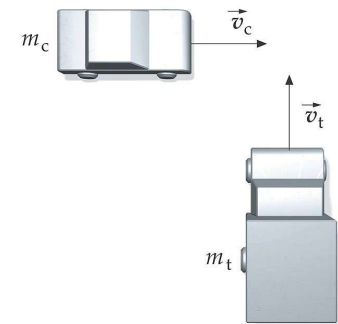
$$\vec{J} \equiv \int_{t_i}^{t_f} \vec{F} dt$$



Totally Inelastic

$$\vec{v}_{f1} = \vec{v}_{f2} \equiv \vec{v}_f$$

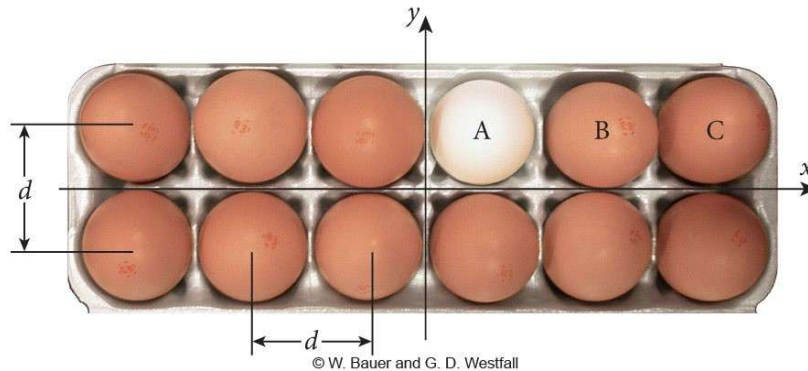
$$\vec{v}_f = \frac{m_1\vec{v}_{i1} + m_2\vec{v}_{i2}}{m_1 + m_2}$$



Summary: Center of Mass

The center of mass of an object is defined as the point at which we can imagine all the mass of an object to be concentrated.

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$$\vec{P} = M\vec{V} = \sum_{i=1}^n \vec{p}_i$$

$$\frac{d}{dt} \vec{P} = \vec{F}_{\text{net}}$$

The forces act upon CM. Only external forces contribute to the net force.

$$\vec{R} = \frac{1}{M} \int_V \vec{r} \rho(\vec{r}) dV$$

